

Exploring Avatar Seat Allocation Strategies in Social AR

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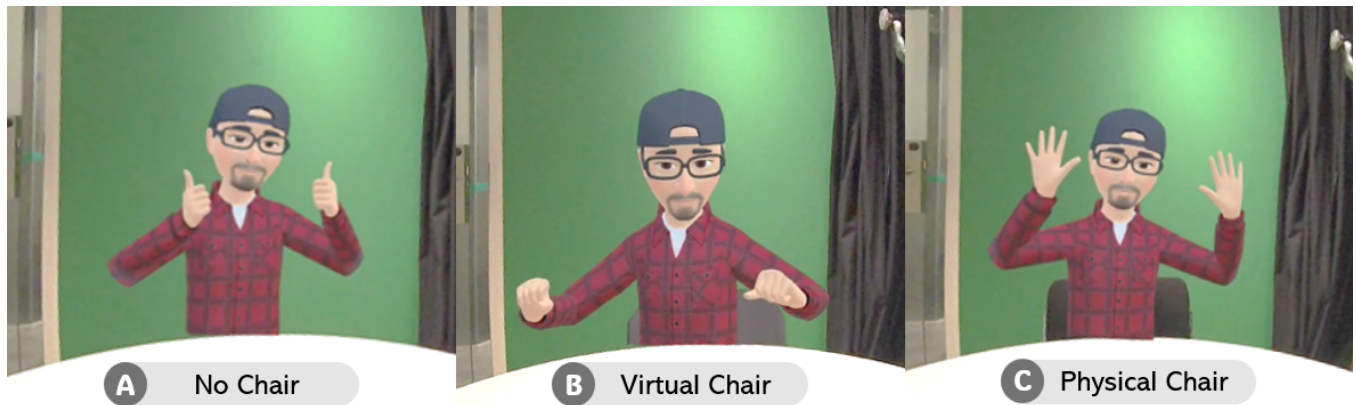


Figure 1: Our study compared three ways to represent virtual avatars in the physical environments of multi-user AR platforms: A) without any chair, B) on a virtual chair, C) on a physical chair. Participants were asked to complete a think-aloud task where they indicated their preferred avatar visualisation in 7 different social scenarios of varying quantities of physically present people, avatars and physical chairs available.

Abstract

In augmented reality (AR) collaboration, the visual representation of remote users' avatars contributes significantly in fostering social presence, immersion, and effective communication. We conducted a within-subjects study where a seated participant interacted with a remote actor under three conditions: the avatar seated on a virtual chair, seated on a physical chair, and not seated. Participants completed a think-aloud exercise to evaluate avatar visualization preferences across varied social scenarios. We found participants overwhelmingly preferred seated avatars, but their allocation to physical or virtual chair was highly scenario-dependent, being

affected by the amount of space available, of avatars in the meeting, and of physical chairs available. We identify three main allocation strategies, prioritising either physical chair occupancy, avatar seat consistency, or virtual chairs for the avatars. We conclude with design guidelines for future AR teleconferencing systems.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; *User studies*; **Mixed / augmented reality**.

Keywords

Augmented Reality, Mixed Reality, Collaboration, Context-Awareness, Affordance, Hybrid Work, Think-Aloud, Thematic Analysis

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1 Introduction and Background

The metaverse is a highly immersive, embodied form of the Internet, revolutionizing how we work, socialize, and interact remotely, moving beyond the constraints of traditional 2D experiences [44]. Devices like Augmented Reality (AR) glasses bring this vision to life in an everyday capacity [5, 32, 37], overlaying virtual elements onto users' physical surroundings to create interactive and engaging environments [14, 20]. Despite their potential, the interplay between virtual content and the physical world remains complex and poorly understood—particularly in remote collaboration, where remote users are represented as avatars within the local user's physical space. For example, little research has explored where to best position avatars in augmented environments, and how and whether they should be anchored or not to the physical space around the user.

Norman defines an affordance as an "*action possibility* that is readily perceivable by an actor" [30] determining how an object may be used or interpreted by an individual: for example, a chair affords the act of sitting, while a whiteboard does not. Similarly, elements within a physical environment may serve as compelling placement options for the avatars of remote collaborators. Imagine a user in their office working with a remote colleague via AR glasses. Do physical chairs function as affordances for avatar placement? Should the avatar 'sit' on an available physical chair at the desk, spawn its own virtual chair to 'sit' on, or neither?

As AR devices and the metaverse become integrated into daily life [32, 37], the interaction between virtual content and the physical world will profoundly influence the design of these technologies. At the same time, how AR technologies are designed will also shape how individuals will behave in daily life and engage with these systems in real-world scenarios. Understanding user preferences for avatar placement within physical environments is essential for designing systems that feel realistic, socially acceptable, and immersive, ultimately shaping the future of remote collaboration. Prior research emphasized that AR teleconferencing systems must be aware of users' physical contexts since remote collaborators interacting realistically with the environment improves co-presence [7]. Additional research has highlighted the importance of realistic AR experiences [1, 15, 16, 19, 21, 41, 42, 42, 43], e.g. respecting the personal space of avatars or human-embodied agents as if physically present humans [9, 12, 27, 29, 45]. Users prefer interacting with virtual avatars that adhere to realistic behaviours by *aligning* with elements of their physical setting while *avoiding* "physicality conflicts" - situations where virtual and real objects occupy the same space [4, 18, 22, 24–26, 31]. These studies show that avatars which respect physical rules or realistically react to their environment, even when physicality conflicts are unavoidable, enhance user experience and improve co-presence and social presence [7, 18, 26, 31, 34]. However, despite these findings, the question of how they translate specifically to avatar placement within the physical context remains unresolved.

Previous research on collaborative AR platforms has explored various approaches to avatar representation. Systems have been

developed, for example, to allow remote collaborators' avatars to appear seated on the chairs present in the physical environment [13, 35, 46]. Pejisa *et al.* [38] extended this concept by enabling avatars to "sit" on either physical chairs or virtual chairs spawned *ad hoc*. Despite these advancements, a critical question remains unanswered: do users sitting at a table prefer the avatars of their remote collaborators to be anchored to chairs, and if so, are physical chairs more effective anchors than virtual ones? Furthermore, how should collaborative AR platforms allocate seats to multiple avatars if there are limited physical chairs available in the local user's context? Addressing this gap is essential for understanding user preferences and designing telepresence systems that enhance user experience and social acceptability in AR collaborative platforms.

We conducted a within-subjects study (N=24) to investigate the impact of different avatar visualisations in AR environments and to explore emerging social dynamics in hybrid-distributed collaboration. In the first stage of the study, a seated participant engaged in a table-based interaction with a remote confederate using a custom AR platform modelled after established collaborative XR systems, such as Meta Horizon Workrooms¹. The participants experienced three conditions: the avatar seated on a physical chair, seated on a virtual chair, and without any chair. Participants were then asked to conduct a think-aloud task to investigate how they would represent remote colleagues in 7 social scenarios where the number of physical co-attendants, remote users, and available physical chairs varied. Finally, we closed by capturing participants general preferences towards our avatar visualisations and reflections about their use.

Participants overwhelmingly favoured seated avatars, mentioning greater social presence and immersion with avatars seated on virtual or physical chairs. Allocation of physical chairs was desired when surplus chairs were available, with participants mentioning a dependence on chairs allocated to local (physically present) users. Some participants preferred virtual chairs for avatars to ensure equal social importance and visual consistency between remote users or to maintain a clear distinction between physical and virtual elements. Strategies for seating avatars varied, including prioritizing physical chair occupancy and prioritizing avatar placement consistency.

We contribute:

- An empirical evaluation comparing AR collaboration with remote users represented as avatars seated on physical chairs, virtual chairs, or not seated at all, finding a clear preference for both seated conditions over the non-seated one, highlighting the importance of seated avatar rendering for improved user experience and social cohesion in distributed AR environments;
- A set of strategies adopted by participants when deciding how to best place remote users' avatars according to their preferences;
- A set of design recommendations highlighting the need for dynamic, personalised, context-aware avatar placement algorithms.

¹<https://forwork.meta.com/gb/horizon-workrooms/>

2 Methodology

2.1 Apparatus

The study utilized Meta Quest 3 headsets running a collaborative app developed in Unity which allowed participants to converse remotely with an actor. Remote users were embodied by choosing one of a set of 32 pre-designed Meta avatars² featuring body and lip tracking. Photon Fusion³ allowed to establish connectivity over WiFi between the apps. The app was designed to replicate the user experience of established XR collaborative platforms such as Meta Horizon Workrooms⁴ and Microsoft Mesh⁵. Participants sat at a round table with a 1.2m diameter allowing comfortable social distances with collaborators [2, 3, 8–11, 23, 28], while the actor was located in a separate room.

2.2 Procedure

After signing consent forms and completing demographic surveys, each participant was asked to use the headset, select their avatar, and start collaborating with the actor, who was (physically) located in a different room and collaborating via the same application. The participant and the actor were asked to conduct a casual conversation about collaboration in AR and what it would look like, and how the custom app compared to their expectations, tracing prior work [33, 39]. The study involved three avatar anchorage conditions, where the avatar of the confederate was either rendered to be sitting on a virtual chair spawned below it (see Fig.1B), on a physical chair (see Fig.1C), or neither (see Fig.1A). The model for the virtual chair was designed to be as similar to the physical chair as possible, as shown in Figure1B and 1C.

After experiencing all of the conditions, we conducted our think-aloud task to explore how preferences for using our avatar representations across 7 social scenarios, tracing O'Hagan's study procedure [34], to gather information regarding the social dynamics that would arise in AR co-located and/or distributed meetings. For more details on this see Section 2.2.1. Finally, we closed our study by asking participants to reflect on our avatar visualisations and to rank order them by preference and justify their ordering. We also asked participants, if they were the remote colleague, how they would prefer to be visualised as an avatar, if they would prefer to have control over their placement within the room, and if they were an avatar seated on a physical chair how they would feel if a local user joined the meeting and took their physical chair. Each study lasted at most one hour, and participants received a £10 Amazon voucher for their participation.

2.2.1 Our Think-aloud Task: Our think-aloud task consisted of 7 static, social scenarios that varied by the number of local and remote participants and physical chairs in the room. For example, a 2-2-4 setup with 2 local users, 2 remote users, and 4 chairs would ask the participant to *"imagine you (physically present) are about to have a meeting with 1 local colleague (also physically present), 2 remote colleague (an avatar) and there are 4 physical chairs at the table"*. The selection of seven different scenarios allowed us

to capture diverse meeting configurations relevant to real-world collaboration use cases. These scenarios included contexts with varying numbers of local and remote colleagues, reflecting common workplace dynamics where teams may have more, equal, or fewer remote members compared to local ones. Additionally, the scenarios considered the availability of physical chairs, which could be more, equal to, or less than the total number of participants. By examining these variations, we aimed to understand how participants balance the needs of both local and remote users in practical settings, such as hybrid work environments.

For each scenario participants were tasked with assigning a preferred seat type (physical, virtual or none) to each user individual in the scenario (including themselves). Participants were instructed to think-aloud and justify their assignments while completing the task. Our 7 scenarios are summarized in Table 1 alongside the quantitative results of participants assignments.

3 Results

In this late-breaking work, we present a subset of our findings, focusing on the qualitative think-aloud data and the quantitative ranking data collected during the study, with future work aiming to delve into the complete dataset. These findings highlight user preferences and social dynamics emerged from the experiment.

Participants: We recruited 24 participants (mean age= 24.04, SD= 5.48, female = 6) via university mailing lists, with varied familiarity with AR interfaces (63% never/rarely, 38% at least weekly) and videoconferencing platforms (42% never/rarely, 58% at least weekly). Most participants felt of a similar social status as the actor (71% equal, 12% similar).

3.1 Think-aloud Results

We conducted an inductive, data-driven thematic analysis of the audio-recorded think-aloud data from the survey. We employed an open coding approach, followed by axial coding, to identify and categorise relevant themes and patterns within the interview recordings [40]. These were auto-transcribed using WhisperX⁶, reviewed, and anonymised. Two researchers familiarised themselves with the dataset, independently created coding schemes using NVivo⁷ and analysing the entire dataset, and then compared, refined, and merged their coding schemes. After developing higher-level codes, they iteratively discussed and integrated these codes into the following themes.

3.1.1 Locals Need and Deserve Physical Chairs: All participants ($N = 24$) agreed that physical chairs should be prioritized for local users. As avatars do not need chairs to sit on, local users should logically take the chairs. Most ($N = 20$) emphasised that locals *"should take priority over the Avatar on who gets to sit on a physical chair"* (P17). As such, in the scenarios like 2-1-2, where there are only as many physical chairs as local colleagues, most participants ($N = 23$) believed that *"virtual chairs are a very nice alternative if you do not have enough chairs in the real world"* (P4). Some participants ascribed a sense of "value" to physical chairs, perceiving them as resources more "deserving" for locals (P24: "I

²<https://www.meta.com/gb/avatars/>

³<https://doc.photonengine.com/fusion/v1/fusion-intro>

⁴<https://forwork.meta.com/gb/horizon-workrooms/>

⁵<https://www.microsoft.com/en-us/microsoft-teams/microsoft-mesh>

⁶<https://github.com/m-bain/whisperX>

⁷<https://lumivero.com/products/nvivo/>

Scenario	Selection	Count
1-1-2	You-PC, Remote-PC	13
	You-PC, Remote-VC	9
	You-PC, Remote-NC	2
2-1-3	You-PC, Local-PC, Remote-PC	16
	You-PC, Local-PC, Remote-VC	7
	You-PC, Local-PC, Remote-NC	1
2-1-2	You-PC, Local-PC, Remote-VC	23
	You-PC, Local-PC, Remote-NC	1
2-2-4	You-PC, Local-PC, Remote-PC, Remote-PC	16
	You-PC, Local-PC, Remote-VC, Remote-VC	7
	You-PC, Local-PC, Remote-NC, Remote-NC	1
2-2-3	You-PC, Local-PC, Remote-VC, Remote-VC	16
	You-PC, Local-PC, Remote-VC, Remote-PC	5
	You-PC, Local-PC, Remote-PC, Remote-NC	1
	You-PC, Local-PC, Remote-NC, Remote-NC	1
	You-NC, Local-PC, Remote-PC, Remote-PC	1
3-1-2	You-NC, Local-PC, Local-PC, Remote-VC	14
	You-NC, Local-PC, Local-PC, Remote-NC	4
	You-PC, Local-PC, Local-NC, Remote-VC	3
	You-NC, Local-NC, Local-NC, Remote-NC	2
	You-NC, Local-NC, Local-NC, Remote-VC	1
1-3-2	You-PC, Remote-VC, Remote-VC, Remote-VC	16
	You-PC, Remote-VC, Remote-VC, Remote-PC	6
	You-PC, Remote-VC, Remote-PC, Remote-NC	1
	You-PC, Remote-NC, Remote-NC, Remote-NC	1

Table 1: The table presents participants' preferred seat allocations for each social scenario, alongside the frequency of each selected allocation set. Each scenario is characterized by three numbers: (1) the number of local users physically present (including the participant, who imagines themselves in the scenario), (2) the number of remote users embodied by virtual avatars, and (3) the number of physical chairs available at the table. Allocations are specified by participant type - You/Local/Remote - and the seat type assigned - No Chair (NC)/Virtual Chair (VC)/Physical Chair (PC). For example, "You-PC" indicates that the participant selected a physical chair for themselves.

think I should have the physical chair because I deserve it"), reflecting a distinct social dynamic: "People would see virtual people are less important than the physical people, like they're a lower priority" (P17).

3.1.2 Social Dynamics of Physical Chairs: Thus physical chairs were perceived as valuable resources, with avatars "managing" to occupy them seen as particularly socially important, especially if locals were left standing: "Making somebody stand in deference to virtual avatars is a bit much, unless that was the dean, the king, or something" (P2). Because of this, participants expressed frustration at scenarios where avatars took the last physical chair (P15: "people would get really angry if they're showing up to the meeting late and someone who's not even there has a chair") and believed avatars should offer their seat to local colleagues: "I would hope that the Avatar would ask: «Do you want this chair?» It's kind of like an old person with a young person, respect your real people" (P14). Some

participants mentioned they would give up their physical chair to an avatar only if it embodied a remote user of high social status, albeit reluctantly: "If the virtual person has a higher social status than me, I'm not going to ask them to move. I'd probably be passive-aggressive about it" (P20). As remote users, many ($N = 11$) claimed they would feel understanding if locals claimed their avatar's physical chair, especially when asked first ($N = 6$, P5: "If they said: «Simon is going to use your chair because it's the only physical chair here and we're going to move you» I'd be like: cool"). However, P13 felt that moving an avatar from a physical to a virtual chair could undermine their perceived importance: "I would feel a little offended, like I've just been made less important. I'd think people might take me less seriously".

3.1.3 The Empty Chair Problem: While everyone agreed physical chairs should first and foremost be given to locals, a key topic of divergence in participant opinion was where to place the avatars in cases with spare physical chairs at the meeting table. Three strategies emerged to deal with the scenario: prioritising filling up all the physical chairs, prioritising consistency in seat type between avatars, and prioritising giving virtual chairs to avatars.

Occupancy-over-Consistency: Many participants ($N = 17$) believed that in at least one social scenario all physical chairs should be occupied before assigning avatars to virtual chairs: "I would use the physical chairs. They're there, let's use them" (P8). This strategy was common in scenarios with enough spare chairs for every avatar, such as 1-1-2, 2-1-3, or 2-2-4, where 13, 16, and 16 participants adopted it, respectively. It was also applied in scenarios 2-2-3 and 1-3-2, where there is one available chair and more than one avatar, with 7 participants using it each. P11 claimed they would rather stand "than seeing [the physical chairs] empty. I would choose the avatars to sit on them". Some participants prioritized avoiding empty chairs for aesthetic reasons ($N = 7$, P6: "It feels awkward if one of the chairs is empty, it gives me the impression that someone's missing") or to reduce visual clutter ($N = 4$, P1: "If there's chairs already you may as well use them, even if the person's not physically there, whereas, if you were to add virtual chairs in, I think it could get cluttered quite easily"). Others found it immersive and conducive to efficiency: "get on with the meeting faster if I was in a work situation" (P21). Interestingly, a few participants ($N = 3$) chose to fill all physical chairs despite generally preferring virtual chairs: "The virtual chair will always be number one because of its potential but if there was a physical chair present I would want to fill it" (P6).

Consistency-over-Occupancy: Many participants ($N = 15$) mentioned for at least one social scenario that they preferred avatars to all use the same type of seat rather than having "one remote person on a real chair while the other didn't have a real chair. I'd make them symmetrical and give them both virtual chairs" (P18). This strategy was common in scenarios with one available chair and more than one avatars, such as 2-2-3 and 1-3-2, each with 17 participants adopting it. Motivations were mainly social or aesthetic. Some ($N = 5$) found mixed-seat-type allocations visually distracting: "It might be more distracting if one of them is on a real chair and the other on a virtual one" (P17). Others ($N = 13$) mentioned it felt "unfair to have one have a real seat and the others have virtual ones. Who do you choose gets the real seat? They would be othered almost" (P22) and that "to maintain equality, [they would rather] leave a

	1st	2nd	3rd	Average Ranking
Physical Chair	13	10	1	2.50
Virtual Chair	10	13	1	2.38
No Chair	1	1	22	1.12

Figure 2: The ranking distributions and average ranking scores (out of a possible 3.00) for our conditions. These results show a strong preference for the seated avatar visualisations.

physical chair empty and have them both in virtual chairs" (P2), as otherwise the avatars would be perceived as having differing social status. Some ($N = 4$) adopted this strategy even when they generally preferred physical chairs: "What would make the most sense is for everyone to be in a physical chair, as if part of the same group, but then that might separate them. You might take the avatar in the physical chair more seriously than those in the virtual chair. So... they should all be in a virtual chair, to kind of put them on an equal level" (P13).

Virtual Chairs to Virtual People: A few participants ($N = 7$) consistently preferred giving avatars virtual chairs. Some preferred to draw a visual distinction between remote and local colleagues ($N = 5$, P7: "I'd like to maintain the virtual world and the physical world separate"), while others did not like seeing virtual avatars on physical chairs ($N = 8$, P20: "A physical chair looks more real than a virtual person sitting in it, drawing attention to the fact that they aren't as real"). This strategy resulted in 9 preferring virtual chairs in the 1-1-2 scenario, and 7 in the 2-1-3 and 2-2-4 ones,

3.1.4 Avatars Should Have Chairs: Most people ($N = 23$) preferred having avatars sit down either in physical or virtual chairs: "I just want everyone to be sitting on chairs: it just feels more like a meeting when everyone's sitting around" (P3). Whether participants found it disrespectful towards the avatar ($N = 7$, P18: "It's like you were squatting. It gives the aura that I'm not valued enough to get a seat."), or found it negatively impacted their sense of immersion ($N = 12$, P8: "It makes it more immersed, rather than floating in the environment, it feels more real"), only one participant consistently preferred interacting with avatars that were not sitting on a chair.

3.2 Rank Ordering of Avatar Visualisation Results

Figure 2 summarises the user preference distributions and average ranking scores for avatar seating visualisations. A Friedman test with Bonferroni-corrected p -values indicated significant differences between participants' preferred avatar visualisation ($\chi^2 = 27.75$, $p=0.017$). The effect size of the Friedman test was calculated using Kendall's W ($\eta^2 = 0.064$) indicating a small effect size, which makes sense given the differences in preference seen in Figure 2. That is, for nearly all participants, a ranking *Physical Chair* and *Virtual Chair* as either first or second respectively and then ranking *No Chair* third. We analysed pairwise comparisons using Wilcoxon Signed Rank tests with effect sizes for statistically significant comparisons using rank-biserial correlation (r). We found significant differences and large effect sizes between *No Chair* - *Physical Chair* ($p=0.017$, $r=0.827$), and between *No Chair* - *Virtual Chair* ($p=0.017$, $r=0.783$),

indicating users strongly preferred interacting with avatars that were shown to be sitting in virtual and physical chairs rather than avatars that were not shown to be sitting on anything.

4 Limitations

While our think-aloud method provides rich, valuable qualitative insights by allowing participants to verbalize their decision-making processes, it also presents a methodological limitation: the challenge of perspective-taking. Participants must imagine themselves in various scenarios and anticipate their feelings, which may not fully capture real-world nuances, potentially leading to inaccurate preferences. Nonetheless, this explorative think-aloud approach, increasingly used in recent literature [6, 34, 36], offers a deeper understanding of the motivations behind avatar seating choices, contributing to the development of user-centric AR systems.

5 Discussion and Future Work

Our findings reveal important dynamics in how participants assigned physical and virtual seating to avatars and local users in AR multi-user platforms. By examining these results, we contribute to the understanding of how physical resources and social dynamics influence user preferences on avatar placement and rendering in augmented reality environments. Participants strongly preferred interacting with remote avatars seated in either virtual or physical chairs, consistently ranking the No-Chair visualization as their least favourite. This aligns with prior research showing user preferences for avatar behaviour that mimics the behavioural norms of physically present humans [4, 7, 17, 22, 24, 25, 29]. These findings suggest that while prior studies have demonstrated the feasibility of automatically placing avatars in physical spaces [13, 35, 46], virtual chairs - despite limited exploration in recent literature [38] - offer a valuable alternative for rendering and positioning remote users. However, the preference for physical versus virtual chairs was context-dependent. Avatars in virtual chairs were seen as wasteful when physical chairs were available, raising concerns about unused resources and social equality. Conversely, avatars occupying physical chairs while local users stood were deemed inappropriate, reflecting a shifting perception of the symbolic and practical value of physical seating. These nuances underscore the potential for innovative research aimed at enhancing the user experience of AR platforms and to design smarter, more context-aware collaborative multi-user systems. This issue emphasizes the practical and symbolic role of physical resources in AR collaboration, expanding on prior work that highlights the influence of physical artifacts on collaborative AR interfaces [4, 22, 24, 25]. Addressing this challenge requires intelligent platforms capable of balancing aesthetic preferences, social expectations, and functional demands. By doing so, AR systems can better manage physical and virtual seating, but further research is required to understand how these changes foster or impact social equity and improve user experience in distributed augmented-reality environments.

5.0.1 Implications for AR Multi-User Platform Design: Our findings suggest several practical recommendations for AR platform developers:

- (1) **Seated Avatar Placements or Renderings:** AR collaboration tools to consistently position remote users' avatars on

physical chairs within the environment or provide virtual chairs, as it was shown from the qualitative data and the quantitative ranking data that "there is no scenario where the avatar shouldn't be sat down" (P15).

- (2) **Dynamic Seating Assignment:** AR platforms should incorporate intelligent algorithms to dynamically allocate physical and virtual chairs based on user roles, presence, and social context. The system should both place avatars on available physical chairs while prioritizing local user allocation - reducing visual clutter and avoiding the awkwardness of unused chairs in the environment - and spawn virtual chairs for avatars when the number of participants exceeds the available physical seating, reducing inequity by preventing "any kind of weird chair-based hierarchy forming [...] and keeping things more visually consistent" (P5).
- (3) **Visual Indicators of Chair Allocation:** To prevent misunderstandings or perceived disrespect, platforms should provide clear visual indicators of how chairs are assigned and occupied. For instance, avatars could "offer" their physical chairs to locals when seating becomes constrained, reinforcing social norms.
- (4) **Customizable Seating Policies:** Recognizing that user preferences vary, AR platforms should allow users to customize seating arrangements based on their priorities, whether that's maximizing physical chair usage or maintaining consistency in seating types.

6 Conclusion

We conducted a within-subjects user study comparing three avatar visualisations (on a virtual chair, on a physical chair or without either) and investigating the resulting social dynamics in AR teleconferencing platforms. Participants preferred seated avatars, expressing nuanced strategies for allocating physical and virtual chairs, either prioritizing occupying as many physical chairs as possible, or keeping every avatar on the same seat type. These were not based on mere logistical concerns but also on deeply social ones, tied to perceptions of presence, equity, and group cohesion. Based on these findings we developed novel guidelines for more inclusive and context-aware AR teleconferencing platform design, enhancing the quality of hybrid collaboration.

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